1.5.1 Type of Relations

Consider the following examples:

(i) Let $S = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (1, 3), (2, 3)\}$ on S.
(ii) Let $S = \{1, 2, 3, \ldots 10\}$ and define "m is related to n, if m divides n".
(iii) Let $\mathscr{C}$ be the set of all circles in a plane and define "a circle C is related to a circle C', if the radius of C is equal to the radius of C'".
(iv) In the set S of all people define "a is related to b, if a is a brother of b".
(v) Let S be the set of all people. Define the relation on S by the rule "mother of".

In the second example, as every number divides itself, "a is related a for all $a \in S$"; the same is true in the third relation also. In the first example "a is related a for all $a \in S$" is not true as 2 is not related to 2.

It is easy to see that the property "if a is related to b, then b is related to a" is true in the third but not in the second.

It is easy to see that the property "if a is related to b and b is related to c, then a is related to c" is true in the second and third relations but not in the fifth.

These properties, together with some more properties are very much studied in mathematical structures. Let us define them now.

Definition 1.3

Let S be any non-empty set. Let R be a relation on S. Then

- R is said to be ***reflexive*** if a is related to a for all $a \in S$.
- R is said to be ***symmetric*** if a is related to b implies that b is related to a.
- R is said to be ***transitive*** if "a is related to b and b is related to c" implies that a is related to c.

These three relations are called ***basic relations***.

Let us rewrite the definitions of these basic relations in a different form:

Let S be any non-empty set. Let R be a relation on S. Then R is

- reflexive if "$(a, a) \in R$ for all $a \in S$".
- symmetric if "$(a, b) \in R \Rightarrow (b, a) \in R$".
- transitive if "$(a, b), (b, c) \in R \Rightarrow (a, c) \in R$".

Definition 1.4

Let S be any set. A relation on S is said to be an ***equivalence relation*** if it is reflexive, symmetric and transitive.

Let us consider the following two relations.

(1) In the set S_1 of all people, define a relation R_1 by the rule: "a is related to b, if a is a brother of b".
(2) In the set S_2 of all males, define a relation R_2 by the rule: "a is related to b, if a is a brother of b".

The rules that define the relations on S_1 and S_2 are the same. But the sets are not same. R_1 is not a symmetric relation on S_1 whereas R_2 is a symmetric relation on S_2. This shows that not only the rule defining the relation is important, the set on which the relation is defined, is also important. So whenever one considers a relation, both the relation as well as the set on which the relation is defined have to be given explicitly. Note that the relation $\{(1, 1), (2, 2), (3, 3), (1, 2)\}$ is reflexive if it is defined on the set $\{1, 2, 3\}$; it is not reflexive if it is defined on the set $\{1, 2, 3, 4\}$.